Abstract

A draft document that descibes the concept of outdated CSI in my simulation framework.

Important Time Periods

- Symbol Period T_{symbol}: the time needed for the transmission of a single symbol vector.
- Block Period T_{block}: the time needed for the transmission of a block of symbols.
- Coherence Time T_c: the time period during which the channel impulse response remains highly correlated ($R_h(T_c) = \frac{1}{2}$).
- Round Trip Time T_{RTT}: the time period it takes for a signal to travel from the BS to a UT and back. This is equal to twice the propagation delay.

Important Parameters

- Round Trip Time to Coherence Time Ratio $\frac{T_{\text{RTT}}}{T_c}$: This parameter determines the delay on the CSI feedback.
- Symbol Period to Coherence Time Ratio $\frac{T_{\text{symbol}}}{T_c}$ This parameter determines the number of symbol per block.

Precoding Level

The precoding level is determined by the switching rate of precoding. When the precoding scheme is updated with every block of symbols, it is called block-level precoding. On the other hand, when the precoding needs to be updated with every new symbol, it is called symbol-level precoding. Symbol-level precoding can be seen as a special case of block-level precoding, where each block consists of only one symbol.

The advantages and disadvantages of both techniques relative to each other are easy to identify. Block-level precoding consumes only a fraction of the computational energy required to compute the precoding scheme compared to symbol-level precoding. On the other hand, symbol-level precoding makes it possible to adapt the precoding scheme immediately to rapidly changing channel impulse responses.

Up to this point, we have consistently used block-level precoding, and we will continue to do so in what follows.

Outdated CSI

As discussed previously, we adopt block-level precoding. A pilot vector is inserted in each block, allowing the user terminal (UT) to estimate the channel impulse response. This estimate is assumed to be perfect. Once the UT has obtained the channel estimate, it feeds this information back to the base station (BS). As a result, the BS only has access to the channel state information (CSI) after a delay equal to one round-trip time (RTT).

In satellite communication systems, the propagation delay, and hence the round trip time, is non-negligible. Consequently, the CSI available at the BS is inherently outdated, which can significantly impact system performance.

In the following, we analyze the impact of the round-trip-time-to-coherence-time ratio on the block duration and the effective CSI feedback delay.

Round Trip Time > Coherence Time

In the case where the round-trip time (RTT) exceeds the coherence time, the block duration is set equal to the RTT. Selecting a shorter block duration would be ineffective, as it would result in the most recent CSI available at the base station (BS) being identical to that used in the previous precoding computation, thereby yielding the same precoding scheme. Conversely, selecting a longer block duration may be considered if the employed precoding technique is sufficiently robust to accommodate the additional CSI imperfections.
Under these conditions, the CSI feedback delay is equal to the block duration and therefore always exceeds the channel coherence time. Consequently, the CSI available at the BS is effectively uncorrelated with the instantaneous channel state.

As expected, this results in a significant degradation in system performance, since precoding techniques fundamentally rely on accurate channel impulse response information. Given that the available CSI is essentially uncorrelated with the current channel conditions, autoregressive (AR) models are unable to provide meaningful predictions of the true CSI and therefore do not yield performance improvements. In such scenarios, robust precoding techniques may offer a viable solution.

Round Trip Time < Coherence Time

When the round-trip time (RTT) is smaller than the coherence time, the base station (BS) can, in principle, always obtain channel state information (CSI) that remains correlated with the instantaneous channel. In this regime, a trade-off arises between the block duration and the quality of the CSI available at the BS. A longer block duration allows more symbols to be transmitted per block, thereby reducing the computational cost per symbol for precoding. However, it also increases the time interval between consecutive pilot transmissions, resulting in a larger CSI feedback delay and, consequently, a reduced correlation between the available CSI and the instantaneous channel state.

We select the block duration as the difference between the coherence time and the RTT. This choice ensures that the CSI feedback delay does not exceed the coherence time, implying that the precoding scheme for each symbol vector is based on CSI that remains correlated with the current channel state:

$$
\begin{aligned}
&T_{\text{block}} = T_c - T_{\text{RTT}} \\
&T_{\text{RTT}} \leq \text{CSI feedback delay}\,[m_{\text{sv}}] \leq T_c \\
&\mathbb{E}\left[\text{CSI feedback delay}\right] = \frac{1}{2}\left(T_{\text{RTT}} + T_c\right) + \frac{T_{\text{symbol}}}{2}
\end{aligned}
$$

In the extreme case of symbol-level precoding (i.e., $T_{\text{RTT}} = T_c$), the CSI feedback delay is always equal to its minimum value, namely the round-trip time.

It is important to note that the pilot symbols are not always transmitted at the beginning of a block. Instead, they are inserted at time

$$
t_p = t_{\text{block}} + \left(-T_{\text{RTT}} \mod T_{\text{block}}\right),
$$

where t_{block} denotes the start time of the corresponding block. The pilot symbols used to obtain the CSI for the computation of the precoding scheme of the current block originate from the block that is $\lceil T_{\text{RTT}}/T_{\text{block}} \rceil$ blocks prior to the current one. This makes sure that the BS always receives the most recent CSI sample possible.

In this regime, the use of an autoregressive (AR) model to predict the channel impulse response based on outdated CSI can potentially yield performance improvements. The order of the AR model corresponds to the number of past channel impulse response samples used to predict the current value. The more recent the most up-to-date channel estimate available at the BS, the higher the probability of obtaining an accurate prediction. In addition, the availability of multiple past channel estimates that are well correlated with the current CSI further increases the probability of a more accurate prediction.

Given the choice of block duration described above, only the most recently obtained CSI sample remains correlated with the value to be predicted. This is reflected in the following condition:

$$R_h(\max(\text{CSI feedback delay})) \geq 0.5 \iff \max(\text{CSI feedback delay}) \leq T_c$$

A possible approach to increase the order of the AR model and thereby improve prediction accuracy, is to reduce the block duration. To obtain an AR model of order p, the block duration must be divided by p. More generally, a pilot vector is inserted every T_{block}/p seconds, providing channel estimates to the BS at the same rate. The precoding update rate can then be chosen independently of this pilot transmission rate.

Simulation Model

Calculations & Graphs

$$\begin{aligned}
&\mathbb{E}\left[\text{CSI feedback delay}\right] \\
&= \frac{1}{\text{svpb}} \sum_{m_{\text{sv}}=1}^{\text{svpb}} \text{CSI feedback delay}\left[m_{\text{sv}}\right] \\
&= \frac{1}{\text{svpb}} \sum_{m_{\text{sv}}=1}^{\text{svpb}} \left(\text{CSI feedback delay}\left[0\right] + m_{\text{sv}}\, T_{\text{symbol}}\right) \\
&= \frac{1}{\text{svpb}} \left(\sum_{m_{\text{sv}}=1}^{\text{svpb}} T_{\text{RTT}} + \sum_{m_{\text{sv}}=1}^{\text{svpb}} m_{\text{sv}}\, T_{\text{symbol}} \right) \\
&= T_{\text{RTT}} + T_{\text{symbol}} \frac{1}{\text{svpb}} \sum_{m_{\text{sv}}=1}^{\text{svpb}} m_{\text{sv}} \\
&= T_{\text{RTT}} + T_{\text{symbol}} \frac{1}{\text{svpb}} \frac{\text{svpb}\left(\text{svpb}+1\right)}{2} \\
&= T_{\text{RTT}} + \frac{1}{2}\,\text{svpb}\, T_{\text{symbol}} + \frac{1}{2}\, T_{\text{symbol}} \\
&\qquad \bullet \quad \text{svpb} = \frac{T_{\text{block}}}{T_{\text{symbol}}} = \frac{T_c - T_{\text{RTT}}}{T_{\text{symbol}}} \\
&= T_{\text{RTT}} + \frac{1}{2}\left(T_c - T_{\text{RTT}}\right) + \frac{1}{2}\, T_{\text{symbol}} \\
&= \frac{1}{2}\left(T_{\text{RTT}} + T_c + T_{\text{symbol}}\right)
\end{aligned}$$